

Answer Key & Teacher Notes

Session 1 – From Table to Graph • Session 2 – Analyzing a Graph

HOW TO USE THIS LESSON'S PAGES

one lesson = two sessions

Warm-Up	First 5–8 minutes of Session 1. Key: Unit Warm-Up Key.
Prerequisite Practice A/B	Before the lesson, for students who need the skill the session uses. Key in that folder.
Guided Notes	During the lesson. Students fill in the blanks as you teach; answers below.
Practice Form A	In class, after the lesson. Two pages. The self-check code tells students if they are on track.
Practice Form B	Same skills, new numbers: reteach, retake, small group or absent students.
Exit Ticket	Last 5 minutes. Two per page; cut in half.
Homework Packet	Sheet 1 after Session 1, Sheet 2 after Session 2. Level 1 has the same problems and answers, with a word bank, step frames and larger type for students who need language support.

SESSION 1 • TEACHER NOTES

From Table to Graph

TALK PROMPTS

- Why does the point (0, 0) belong on the dog-walking graph? What does it mean?
- Why must the input come first in an ordered pair?

PARTNER TASK

Babysitting Pay. One partner fills a table for \$8 per hour (1, 2, 3 and 5 hours); the other writes the ordered pairs and plots them. Switch with Party Balloons at 16 balloons per bunch.

WATCH FOR

- Switching x and y in an ordered pair (plotting (75, 3) instead of (3, 75)).
- Using uneven spacing on an axis, so constant-rate points do not line up.
- Expecting every line to start at (0, 0), even when there is a starting fee.

SESSION 2 • TEACHER NOTES

Analyzing a Graph

TALK PROMPTS

- How does the steepness of a graph show the rate between two quantities?
- Why can't you read the y -value of any point as the rate?

PARTNER TASK

Two Pay Graphs. Job A's graph passes through (3, 24) and Job B's through (3, 15); both start at (0, 0). Each partner finds one rate, then together decide whose line is steeper and how far apart the pay is after 5 hours (\$40 and \$25, \$15 apart).

WATCH FOR

- Reading the y -coordinate of a point as the rate (100 steps in 2 minutes read as 100 per minute).
- Dividing x by y instead of y by x .
- Comparing steepness on two graphs drawn with different scales.

ITEM ANALYSIS

depth of knowledge (DOK) for each practice item

Sheet	Standard	1	2	3	4	5	6	7	8	9	10	11
S1 • Form A	6.EE.C.9	1	2	2	2	2	1	2	3	3	1	1
S1 • Form B	6.EE.C.9	1	2	2	2	2	1	2	3	3	1	1
S2 • Form A	6.EE.C.9	1	2	1	2	2	3	2	3	3	1	1
S2 • Form B	6.EE.C.9	1	2	1	2	2	3	2	3	3	1	1

Items before the last three are Practice; then Find the Error (DOK 3) and two Test Practice items.

GUIDED NOTES

ordered pair: Two numbers (x, y) that tell where a **point** is.

coordinates: The **two** numbers in an ordered pair.

axis: One of the two **number lines** on a graph.

plot: To **mark** a point on a graph.

- Each row of a table becomes an ordered pair: x **first**, then y .
- To plot (x, y) , start at $(0, 0)$. Move **right** x , then **up** y .

Example: The rows become **(1, 4)**, **(2, 8)** and **(3, 12)**. Plot **(1, 4)**: right 1, then up 4. Answer: The points line up in a **straight** line.

YOU TRY

1. a. (1, 6) b. (2, 12) c. (4, 24)
2. The three points line up in a straight line through $(0, 0)$.

EXIT TICKET

1. a. (2, 18) b. (5, 0)
2. (1, 2), (3, 6), (4, 8)
3. **B** (straight line)

PRACTICE FORM A

1. a. (3, 6) b. (5, 45) c. (0, 7) d. (8, 1)
2. 9, 18, 27, 45; points (1, 9), (2, 18), (3, 27), (5, 45) line up in a straight line through $(0, 0)$
3. (6, 300): after 6 seconds, the lift has risen 300 feet
4. She switched x and y . The nights (x) come first: (3, 75).
5. 6, 15, 24; points (2, 6), (5, 15), (8, 24) in a straight line
6. (5, 90): 5 rows have 90 seeds
7. a. $y = 5x$ (multiply x by 5) b. a straight line through $(0, 0)$ c. (7, 35)
8. (1, 6), (2, 8), (3, 10); the line meets the y -axis at $(0, 4)$, the \$4 fee
9. He switched x and y . Hours are x and dollars are y , so the points are (3, 15) and (5, 25).
10. **B** $((4, 20))$
11. **C** $((3, 27))$

Self-check word GRAPH

PRACTICE FORM B

1. a. (4, 28) b. (7, 2) c. (0, 5) d. (6, 54)
2. 8, 16, 24, 40; points (1, 8), (2, 16), (3, 24), (5, 40) line up in a straight line through $(0, 0)$
3. (8, 96): after 8 seconds, the elevator has risen 96 feet
4. He switched x and y . The days (x) come first: (3, 45).
5. 16, 48, 80; points (1, 16), (3, 48), (5, 80) in a straight line
6. (6, 84): 6 rows have 84 tiles
7. a. $y = 4x$ (multiply x by 4) b. a straight line through $(0, 0)$ c. (9, 36)
8. (1, 8), (2, 13), (3, 18); the line meets the y -axis at $(0, 3)$, the \$3 fee
9. She switched x and y . Cars are x and dollars are y , so the points are (2, 12) and (5, 30).
10. **C** $((6, 42))$
11. **B** $((5, 40))$

Self-check word POINT

GUIDED NOTES

rate of change: How much y changes when x goes up by 1.

unit rate: The amount of y for **one** unit of x .

origin: The point **(0, 0)**, where the two axes cross.

steeper: A line that rises **faster** as you move right.

- Rate of change = change in $y \div$ change in x .
- If a line goes through $(0, 0)$, divide **y** by x for any point on it.

Example: Change in y : $15 - 6 = 9$ dollars. Change in x : $5 - 2 = 3$ hours. Divide: $9 \div 3 = 3$. Answer: The rate is **\$3** per hour.

YOU TRY

- a. 5 b. 7
- $(20 - 8) \div (5 - 2) = 4$ miles per hour; $4 \times 9 = 36$ miles

EXIT TICKET

- a. 8 b. 6
- $(48 - 24) \div (6 - 3) = \8 per bag; 8 bags: $8 \times 8 = \$64$
- B** (9)

PRACTICE FORM A

- a. 7 b. 5 c. 3 d. 9
- a. 4 minutes cost \$28 b. \$7 per minute c. \$21
- a. 2 miles per hour b. $2 \times 8 = 16$ miles
- a. 4 MB per song b. 5 songs use 20 megabytes
- a. $(28 - 14) \div (4 - 2) = 7$ inches per week b. $7 \times 9 = 63$ inches
- Job B is steeper: Job A pays \$6 per hour and Job B pays \$9 per hour, so Job B pays more each hour.
- a. 25 gallons per hour b. (10, 250)
- (3, 45) means 3 signs raise \$45 in all, not per hour. $\$45 \div 3 = \15 per sign.
- She used the y -value of (2, 100) as the rate but forgot to divide by 2 minutes. $100 \div 2 = 50$, so the rate is 50 steps per minute.
- A** (8 per unit)
- B** (a greater rate)

Self-check word **RATES**

PRACTICE FORM B

- a. 6 b. 4 c. 8 d. 11
- a. 4 hours cost \$24 b. \$6 per hour c. \$18
- a. 3 miles per hour b. $3 \times 7 = 21$ miles
- a. 5 MB per photo b. 6 photos use 30 megabytes
- a. $(12 - 6) \div (4 - 2) = 3$ centimeters per week b. $3 \times 11 = 33$ centimeters
- Job A is steeper: Job A pays \$8 per hour and Job B pays \$5 per hour, so Job A pays more each hour.
- a. 20 liters per hour b. (12, 240)
- (4, 36) means 4 boxes raise \$36 in all, not per box. $\$36 \div 4 = \9 per box.
- He used the y -value of (3, 120) as the rate but forgot to divide by 3 minutes. $120 \div 3 = 40$, so the rate is 40 meters per minute.
- C** (7 per unit)
- B** (P has a greater rate)

Self-check word **STEEP**

HOMEWORK PACKET • ANSWERS

same answers for the core packet and Level 1

SHEET 1 · FROM TABLE TO GRAPH

1. a. (4, 12) b. (0, 0) c. (6, 30) d. (9, 2)
2. \$4, \$8, \$12, \$20; points (1, 4), (2, 8), (3, 12), (5, 20) on a straight line through the origin
3. a. (5, 10) b. After 3 hours, Kofi has hiked 6 miles. c. 2 miles per hour
4. (1, 6), (3, 18), (4, 24). (0, 0) means 0 hours of renting costs \$0.
5. He switched x and y. The x value (hours) comes first: plot (2, 10) and (4, 20).

SHEET 2 · ANALYZING A GRAPH

1. a. After 4 minutes, the pool has 60 gallons. b. 15 gallons per minute c. 135 gallons
2. a. 7 b. 8 c. 12
3. Unit rate: $18 \div 3 = 6$ pages per hour; 7 hours: $6 \times 7 = 42$ pages
4. Job A: $24 \div 4 = \$6$ per hour. Job B: $18 \div 6 = \$3$ per hour. Line A is steeper, so Job A pays more each hour.
5. No. A proportional graph must pass through (0, 0); this one starts at (0, 10). (0, 10) means she had \$10 saved at the start.
6. **C** (7)
7. **B** (Q is steeper; greater rate.)